



Production and MRP: By Products and Additional Quantity

SAP Business One
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THE BEST RUN 

Welcome to the **By-Product and Additional Quantity** course topic.

In order to take this course, you should first complete the **Production Process** training topic.

Objectives



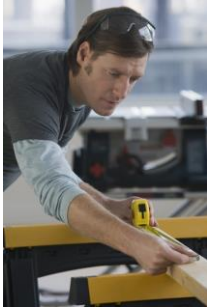
At the end of this topic, you will be able to:

- Explain the concept of a by-product.
- Add By-Products to inventory during the production process
- Define Additional Quantity in the Production Order
- Calculate the planned quantity of items and resources in the Production Order

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- Explain the concept of a by-product.
- Add By-Products to inventory during the production process
- Define Additional Quantity in the Production Order
- Calculate the planned quantity of items and resources in the Production Order

By-Products



What is a by-product?

- A by-product is an item produced from left-over materials in the production process.
- One or more by-products can be produced along with the main product, using a single production order.
- These by-products are then stored in the warehouse as separate items and later are sold or used again as a component in production.

Examples:

- Production of window and door frames, where the remains after cutting are re-used for production of smaller sizes of final products.
- Production of chemicals, where beside the main chemical also some side chemicals can be produced.
- Metal plate cutting, where remains are re-used.

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So What is a by-product?

A by-product is an item produced from left-over materials in the production process.

One or more by-products can be produced along with the main product, using a single production order.

These by-products are then stored in the warehouse as separate items and later are sold or used again as a component in production.

In our business example, production of window and door frames involves cuts that leave wood remains that can be re-used for producing smaller sized items. Other examples include:

Production of chemicals, where beside the main chemical also some side chemicals can be produced.

Metal plate cutting, where the remains are re-used for production.

By-Products in BOM and Production Order

- By-products are defined in BOMs and *Production Orders* as items, but with a negative quantity
- Both the *Manual* and *Backflush* issue methods are supported for by-products

Components		Summary								
#	Type	No.	Description	Base Qty	Planned Qty	UoM Name	Ware...	Issue Method	End Date	
1	Item	→ D000111	Wooden Door	1	20		→ 01	Backflush	▼ 25.10	
2	Item	→ H000001	Handel	2	40		→ 01	Backflush	▼ 25.10	
3	Resource	→ RC0002	Carpenter	1.5	30	hours	→ 01	Backflush	▼ 25.10	
4	Resource	→ L000001	Lathe Machine	1	20	hours	→ 01	Backflush	▼ 25.10	
5	Item	→ BY0001	Wooden bars	-2	-40		→ 01	Backflush	▼ 25.10	
6	Item									

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By-products are defined in Bill of Materials and Production Orders as items, but with a negative quantity.

Both the Manual and Backflush issue method are supported for by-product.

By-products are received to a warehouse by Receipt from Production document, like the main final product.

Here we see a by-product item with a negative quantity of 1 entered on a Production Order row. Since the planned quantity for the finished product is 20, the total of by-products produced during manufacturing will be 40. Because we know that this quantity is always produced when making this finished product, we choose to backflush the by-product.

Receiving By-Products

- By-products are received from production into the warehouse by a *Receipt from Production*, along with the main final product
- When the *Manual Issue* method is used then the quantity can be manually adjusted
- When the *Backflush* issue method is used then the quantity is derived from the quantity of the main final product

#	Order No.	Series No.	Item No.	Item Description	By-Product	Quantity	U.
1	156	Primary	D000123	White Door	☐	20	
2	156	Primary	BY0001	Wooden bars	☒	40	

By-products are received from production into the warehouse by a Receipt from Production, along with the main final product.

When the Manual issue method is used, the quantities can be manually adjusted. However, when the Backflush issue method is used the quantity is derived from the parent item quantity of the main final product..

By-products with Backflush issue method are added or removed together with the parent item, they are bound together.

By-products with Manual issue method can be added or removed from the document as independent lines.

A By-Product column can be added in the Receipt from Production document that indicates if the item is a by-product or not.

Additional Quantities

What is an *Additional Quantity*?

- *Additional Quantity* (Setup Quantity) are items or resources that need to be consumed at the beginning or at the end of a production process.
- *Additional Quantities* are not affected by the number of parent items produced.

Examples:

- Machine startup time (Resource)
- Lead component used to set up the machine (Item)



$$\text{Planned Quantity} = (\text{produced item planned Qty} \times \text{base Qty}) + \text{Additional Qty}$$

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What is Additional Quantity?

Additional Quantity (Setup Quantity) represents items or resources that need to be consumed at the beginning or at the end of a production process.

Examples include:

Machine startup time (Resource).

A lead component used to adjust the machine (Item).

The Planned Quantity of a component in a Production Order is calculated using this formula:

Planned Quantity = (produced item planned Qty x base Qty) + Additional Qty.

This rule applies for both items and resources.

Example: Additional Quantities

Example – Additional Qty represents setup time of the resource

- BOM definition: Qty = 1 , Additional Qty = 0.25
- Planned final product Qty = 10
- Planned resource component Qty = $(10 \times 1) + 0.25 = 10.25$

The screenshot displays two SAP screens. The left screen is the 'Production Order' header, showing details for Product No. D00003, Product Description 'Decorative Door', and Planned Quantity 1. The right screen is the 'Bill of Materials' (BOM) for Product No. D00003, showing a hierarchy of components. The BOM table includes the following data:

#	Type	No.	Description	Quantity	UoM No.	Warehouse	Issue Method	Production Std.	Total Production	Price List
1	Item	D000111	Wooden Door	1	01	01	Backflush	GBP 0.00	GBP 0.00	Base Price
2	Item	H00001	Handle	2	01	01	Backflush	GBP 0.00	GBP 0.00	Base Price
3	Resource	L00001	Lathe Machine	1	01	01	Backflush	GBP 0.00	GBP 0.00	Base Price

Below the BOM table, the 'Components' section shows a detailed view of the 'Lathe Machine' resource component (L00001). The table includes the following data:

#	Type	No.	Description	Base Qty	Additional Qty	Planned Qty	Issued	Available	UoM	Warehouse	Issue M.
1	Item	D000111	Wooden Door	10		10	0	0	Manual	01	Backflush
2	Item	H00001	Handle	20		20	0	0	Manual	01	Backflush
3	Resource	L00001	Lathe Machine	10	0.25	10.25	0	0		01	Backflush
4	Item						0	0			

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Here is an example of calculating an Additional Quantity. Here the Additional Quantity represents the setup time for the resource.

In our business example, we use a lathe machine in the production process.

In the bill of materials we define the Quantity as 1 hour for running the lathe machine, but add an Additional Quantity of 0.25 hours for the time taking to set up the drill of the machine.

In our Production Order we produce a quantity of 10 items, therefore the total time for manufacturing the final product is 10 hours based on 10 times 1 hours. However we must add the additional time for the machine set up. Since the set up is not dependent on the number of items produced, we just add the additional time to the 10 hours manufacturing time and receive a total of 10.25 hours.

Summary



Here are some key points:

- By-products are defined in *Bills of Materials* and *Production Orders* as items, but with negative quantity.
- Additional Quantity is an amount of items or resources that needs to be consumed at the beginning or at the end of a production process, and is not affected by the number of parent items to produce.
- The *Planned Quantity* of a component in the *Production Order* row is equal to:
(the quantity in the row X the planned quantity of the production order) + the additional quantity in the row.

Here are some key points to take away from this session:

By-products are defined in Bills of Materials and Production Orders as items, but with negative quantity.

Additional Quantity is an amount of items or resources that needs to be consumed at the beginning or at the end of a production process, and is not affected by the number of parent items to produce.

The Planned Quantity of a component in the Production Order row equals to:
(the quantity in the row X the planned quantity of the production order) + the additional quantity in the row.

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